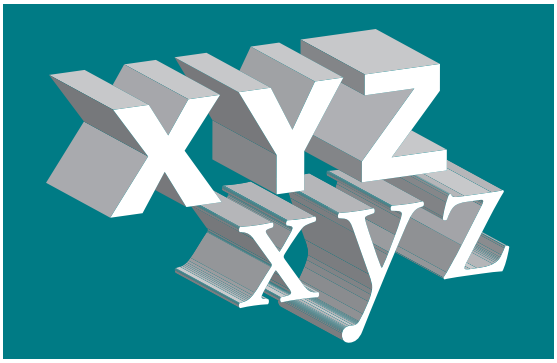


x-axis, *y*-axis

See coordinates

yard

An imperial measure of length.

1 yard = 36 inches ≈ 90 cm

year

The period of time it takes the Earth to make one complete revolution around the sun: 365 days, 5 hours and 48 $\frac{3}{4}$ minutes. The extra hours, minutes and seconds are put together into an extra day every four years to form a 'leap year'.

See day, leap year, revolution

zero

(Symbol: 0, Ø)

The numeral 0 (nought). Nothing.

Rules for working with zero:

1 A number + 0 = same number

$5 + 0 = 5$

2 A number – 0 = same number

$7 - 0 = 7$

3 A number × 0 = 0

$6 \times 0 = 0$

4 0 ÷ any number = 0

$0 \div 10 = 0$

5 A number ÷ 0 has no answer

$3 \div 0 =$ can't do

The digit zero is used as a place holder in numerals.

Example

In the number sixty, 0 is a place holder for units to show that the 6 means six tens and there are no single units.

Note: These words all mean zero: nil, nought, none, nix, null, oh, void, empty set, zilch, duck (in cricket), love (in tennis).

See digit, place holder

zero power

When working with indices, any number raised to the power zero always equals one. It happens because when we divide indices, we subtract the indexes and get zero.

$2^0 = 1 \qquad 376^0 = 1 \qquad x^0 = 1$

Example

Find the value of:

$5^2 \div 5^2$

$= 5 \times 5 \div 5 \times 5$

$= 25 \div 25$

$= 1$

$5^2 \div 5^2$

$= 5^{(2-2)}$

$= 5^0$

$= 1$

See index, index laws, power of a number

